Práctica 3 : Sistema musculoesquelético

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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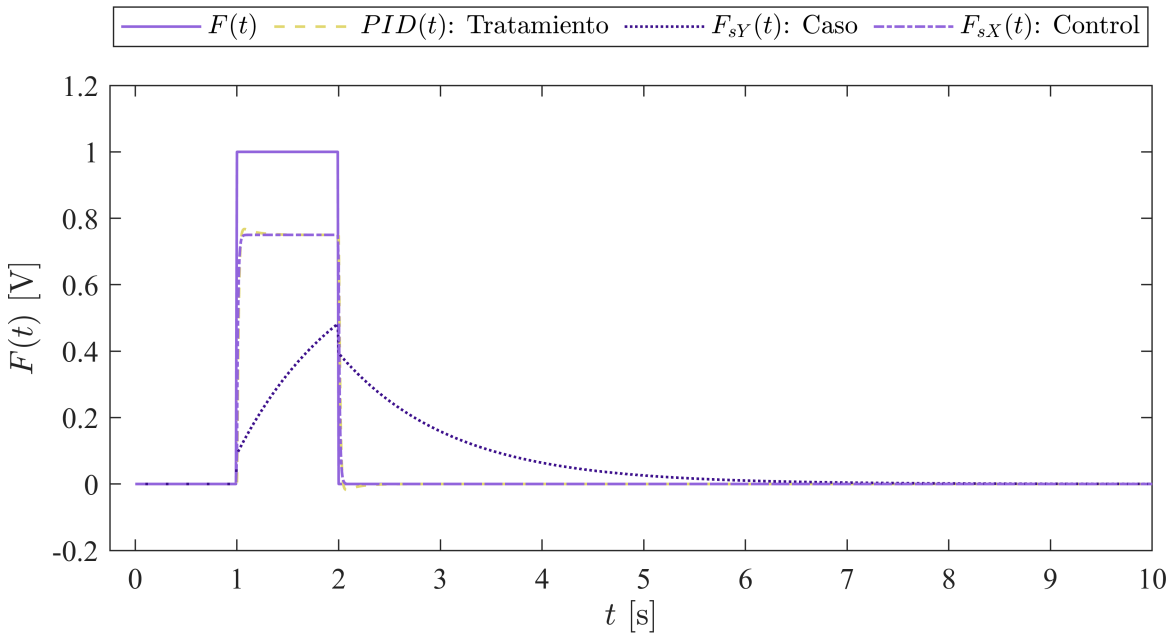
Datos de la simulación

```
clc; clear; close all; warning('off','all')
```

```
open_system('sistema2020_3')
parameters.StopTime = '10';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
x = sim('sistema2020_3',parameters);
```

Respuesta en Caso

```
plotsignals(x.t, x.F, x.PID, x.Fsy, x.Fsx,'Caso')
```



Modelo de EDOs

```
dt = 1e-3;
t = (0:dt:10);
F = zeros(length(t),1); F(1/dt:2/dt) = 1;
PID = interp1(x.t, x.PID, t, 'previous');

par_control = [0.25,10e-6,100e-6,100]
```

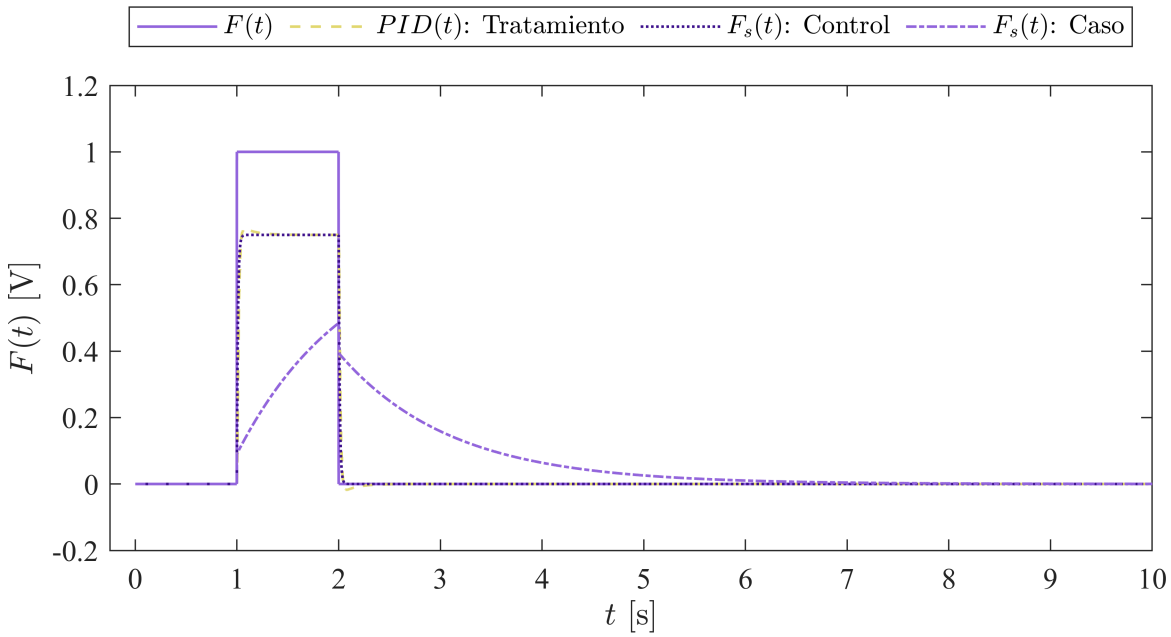
```
par_control = 1×4
    0.2500    0.0000    0.0001  100.0000
```

```
par_caso = [0.25,10e-6,100e-6,10e3]
```

```
par_caso = 1×4
10^4 ×
    0.0000    0.0000    0.0000    1.0000
```

```
Fs_control = musculoesqueletico(par_control,F,t);
Fs_caso = musculoesqueletico(par_caso,F,t);

plotsignals(t,F,PID,Fs_control,Fs_caso,'EDOsys')
```




```
function Fs = musculoesqueletico(par,F,t)
    alpha = par(1); Cs = par(2); Cp = par(3); R = par(4); dt = t(2);
    n = round(t(end)/dt);
    x = zeros(length(t),1); x(1) = 0;
    Fs = zeros(length(t),1); Fs(1) = 0;
    for i = 1:n
        x(i+1) = x(i) +((F(i)-x(i))/(R*(Cp+Cs)))*dt; %EDO
        Fs(i+1) = Cs*F(i)/(Cp+Cs) + (1 - alpha - Cs/(Cp+Cs))*x(i); %Funcion
    end
end
```

Funcion: Respuestas de la simulacion

```
function plotsignals (t,x1,x2,x3,x4,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = ([149, 100, 221])/256;
    clr2 = ([228, 218, 114])/256;
    clr3 = ([62, 15, 141])/256;
    order = [clr1;clr2;clr3]; colororder(order)
```

```
    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$F(t)$ [V]','Interpreter','latex')
    xlim([-0.25 10]); xticks(0:1:10)

    if signal == "Caso"
        plot(t,x1,'-',t,x2,'--',t,x3,':',t,x4,'-.','LineWidth',1)
        L = legend('$F(t)$','$PID(t)$: Tratamiento','$F_{sY}(t)$: Caso','$F_{sX}(t)$: Control')
    elseif signal == "EDOsys"
        plot(t,x1,'-',t,x2,'--',t,x3,':',t,x4,'-.','LineWidth',1)
        L = legend('$F(t)$','$PID(t)$: Tratamiento','$F_{s}(t)$: Control','$F_{s}(t)$: Caso');
    end

    set(L,'interpreter','latex','location','northoutside','orientation','horizontal')
    ylim([-0.2 1.2]); yticks(-0.2:0.2:1.2)

    exportgraphics(gcf,[signal,'_matlab.pdf'],'ContentType','vector')
end
```